

Bridging the Semantic and Visual Gap: A Dual-Model System for Multimodal Fashion Recommendation

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Abstract—The modern fashion e-commerce landscape is defined by two fundamental user needs: finding products that match a specific visual style (visual search) and discovering items from a complex text description (semantic search). This paper presents a robust, dual-pipeline system engineered to master both. Our first pipeline employs a VGG16-based Content-Based Image Retrieval (CBIR) system, leveraging deep visual features to find perceptually similar items. The second pipeline utilizes CLIP, a state-of-the-art contrastive language-image model, to bridge the semantic gap, enabling nuanced, text-to-image retrieval. By building parallel feature indices—a 4096-dimensional space for visual similarity and a 512-dimensional shared embedding space for multimodal search—we demonstrate a comprehensive and practical solution. This hybrid approach moves beyond simple tagging, creating a foundational engine for a truly intelligent and responsive fashion personalization platform.

Index Terms—Fashion Recommendation, Computer Vision, VGG16, CLIP, Multimodal, Deep Learning, Natural Language Processing, CBIR

I. INTRODUCTION

The online fashion industry, a multi-billion dollar market, is not just a storefront; it is an environment for discovery. The central user experience challenge is no longer just inventory, but **discovery**. With vast, ever-changing catalogs, how do users find what they are looking for, especially when they cannot describe it in simple keywords?

This challenge is twofold, mapping directly to two distinct user intentions:

- 1) **Visual Search (Image-to-Image):** A user has an image of a product they love—perhaps from social media or a street-style blog—and wants to find similar items. This is a task of **perceptual matching** based on patterns, colors, and shapes.
- 2) **Semantic Search (Text-to-Image):** A user has an abstract idea, a **feeling**, or a complex need (e.g., “a red floral dress for summer”) and wants to find items that match this description. This is a task of **conceptual matching**.

Traditional search, reliant on manual tagging, fails at both. This paper addresses this gap by engineering a dual-model

system designed to master both modes of discovery.¹ For the visual search task, we leverage the powerful, pre-trained feature extraction capabilities of the VGG16 architecture [1]. For the semantic text-to-image task, we employ the state-of-the-art CLIP model [2], which provides a revolutionary shared embedding space for images and text.

We detail the architecture, indexing process, and retrieval methods for both models, demonstrating their effectiveness as a combined, synergistic solution for the next generation of fashion personalization.

II. BACKGROUND AND RELATED WORK

A. CNNs for Feature Extraction

Content-Based Image Retrieval (CBIR) has long relied on extracting “feature vectors” or “fingerprints” from images. With the rise of deep learning, Convolutional Neural Networks (CNNs) pre-trained on large datasets like ImageNet [3] became the industry standard. Architectures like VGG16 [1] and ResNet [4] have proven to be exceptionally powerful feature extractors. By removing the final classification layers, the output of an intermediate dense layer (e.g., `fc1`) can be used as a high-dimensional vector that encapsulates the visual essence of an image.

B. Multimodal Models (CLIP)

The critical limitation of standard CNNs is their inability to connect images to language. This barrier was broken by multimodal models like CLIP (Contrastive Language-Image Pre-training) [2]. CLIP is trained on hundreds of millions of (image, text) pairs from the internet. It learns to project both images and text descriptions into a shared, 512-dimensional embedding space. In this space, the vector for a photo of a dog is measurably “close” to the vector for the text “a photo of a dog.” This enables a zero-shot, text-to-image search capability without any fine-tuning.

¹The complete source code and implementation files for this project are available at: <https://github.com/Adi-gitX/fashion-recommendation-system>

III. DATASET & PREPROCESSING

For this project, we used the **Fashion Product Images (Small)** dataset, a publicly available collection on Kaggle. This dataset contains thousands of product images from various e-commerce websites, covering a wide range of categories including apparel, footwear, and accessories.

Regardless of the model, a standardized preprocessing step is crucial. All images are first loaded using the Python Imaging Library (PIL) and resized to 224×224 pixels, the standard input resolution for both VGG16 and CLIP. Further model-specific normalization is then applied.

IV. METHODOLOGY: THE DUAL-MODEL SYSTEM

We architected two parallel pipelines that operate on the same set of images but serve different purposes.

A. Model 1: VGG16 for Visual Search (Image-to-Image)

This pipeline is optimized for finding visually identical or near-identical items.

- 1) **Architecture:** We load the VGG16 model pre-trained on ImageNet, removing the final `softmax` classification layer. We create a new Keras model whose output is the `fc1` layer, which produces a 4096-dimensional feature vector.
- 2) **Feature Extraction:** We iterate through every image in the dataset, resize it to 224×224 , and run it through the `preprocess_input` function specific to VGG16. The image is then passed to our modified VGG16 model to extract the 4096-dimensional vector.
- 3) **Indexing:** All extracted feature vectors are stored in a simple NumPy array. For a production system, this would be replaced by a specialized vector database like FAISS.
- 4) **Retrieval:** To find similar items, a query image is processed in the same way. We then use the `NearestNeighbors` (KNN) algorithm from Scikit-learn, configured with 5 neighbors and Euclidean distance, to find the 5 closest feature vectors in the index. The corresponding images are then returned.

B. Model 2: CLIP for Multimodal Search (Text-to-Image)

This pipeline is optimized for semantic search based on natural language.

- 1) **Architecture:** We use the `clip-ViT-B-32` model, loaded via the `sentence-transformers` library. This model is known for its strong balance of speed and performance.
- 2) **Image Indexing:** We iterate through every image in the dataset, load it with PIL, and pass it to the `model.encode()` function. This generates a 512-dimensional embedding for each image. These embeddings are stored.
- 3) **Text-to-Image Retrieval:** When a user provides a text query (e.g., “red floral dress”), the `model.encode()` function is used to convert this text string into its own 512-dimensional vector.



Fig. 1. A query image (left) and the top 5 visually similar results (right) returned by the VGG16 + KNN model.



Fig. 2. The text query “red floral dress” and the top 5 most relevant images retrieved by the CLIP model.

- 4) **Similarity Matching:** Because the text vector and all image vectors exist in the same latent space, we can directly compare them. We calculate the **Cosine Similarity** between the text vector (**T**) and every image vector (**I**) in our index. The images with the highest similarity scores are the most relevant results.

The cosine similarity is calculated as:

$$\text{similarity} = \cos(\theta) = \frac{\mathbf{T} \cdot \mathbf{I}}{\|\mathbf{T}\| \|\mathbf{I}\|}$$

V. RESULTS AND DISCUSSION

The performance of this system is best evaluated qualitatively, as the notebook’s goal is to build the retrieval pipelines.

A. VGG16 Visual Search Results

The VGG16 pipeline excels at its specific task: finding items with near-identical visual properties. As shown in Fig. 1, a query for a specific black and white patterned shirt returns other shirts with highly similar necklines, colors, and pattern densities. This model is extremely effective for “find more like this” or “shop the look” functionality, directly serving a user’s validation intent.

B. CLIP Text-to-Image Results

The CLIP pipeline demonstrates a remarkable, almost human-like, understanding of abstract concepts. As shown in Fig. 2, the query “red floral dress” successfully retrieves images of dresses that are both “red” and “floral,” even as their specific designs, shapes, and lighting conditions vary dramatically. This model is ideal for exploratory search, serving a user’s *discovery* intent.

C. Discussion: A Synergistic System

The two models are not redundant; they are synergistic. They serve two different states of the user’s mind:

- **CLIP is for Exploration:** A user starts with a broad idea. (e.g., “men’s sneakers”).
- **VGG16 is for Validation:** From the CLIP results, the user finds one they like. They click it, and the VGG16 model instantly provides visually similar options.

Together, this dual-model system allows a user to seamlessly transition from broad, abstract discovery to narrow, specific selection, creating a complete and intuitive browsing experience.

VI. CONCLUSION & FUTURE WORK

This paper successfully demonstrates the design and implementation of a dual-model fashion recommendation system. By combining the strengths of VGG16 for precise visual search and CLIP for broad semantic search, we create a flexible and powerful tool for product discovery.

The most significant future work involves the **unification** of this system with other personalization models to create a holistic user profile.

- 1) **Integration with Skin Tone Classification:** The skin tone classification model (developed in our parallel work) can act as a powerful **pre-filter**. A user's skin tone profile (e.g., "Olive") can be used to filter the database **before** the recommendation models retrieve items, ensuring all suggestions are already color-appropriate.
- 2) **Body Shape Analysis:** As planned in our project's next phase, a third model for body shape classification can be added. The final recommendation pipeline would then filter by color (skin tone), style (CLIP/VGG16), and fit (body shape).
- 3) **Scalable Indexing:** For real-world use, the NumPy arrays must be replaced with an approximate nearest-neighbor (ANN) vector index, such as FAISS or Annoy, to enable real-time search on millions of items.

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